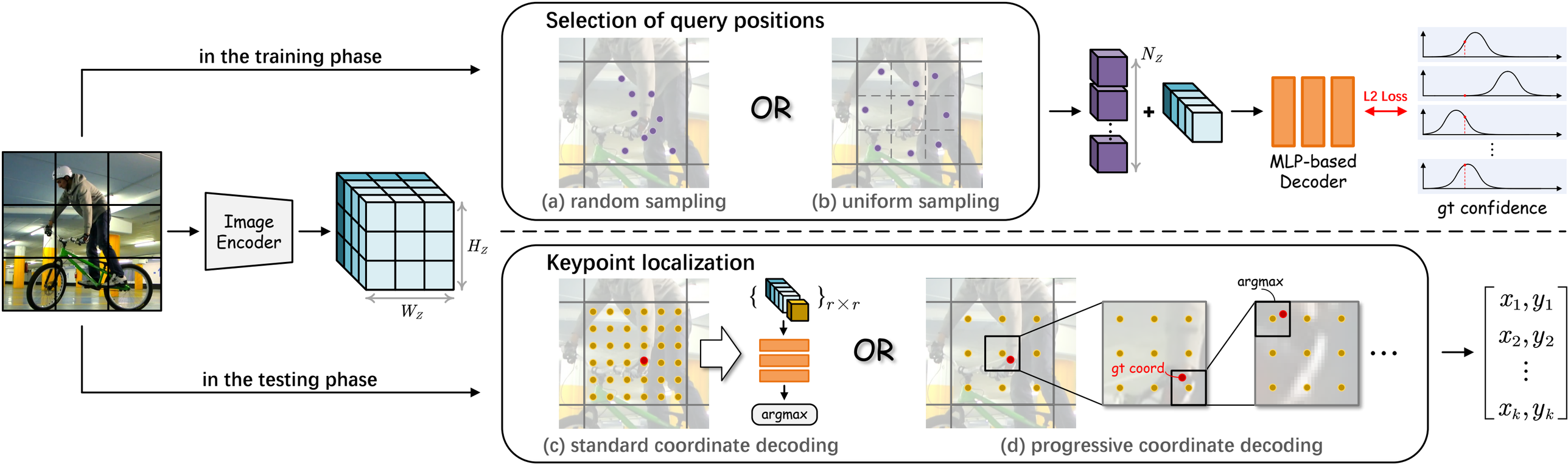
in the training phase
in the testing phase
Image Encoder
H_Z
W_Z
Selection of query positions
OR
(a) random sampling
(b) uniform sampling
N_Z
MLP-based Decoder
L2 Loss
gt confidence
Keypoint localization
$\}_{r \times r}$
argmax
OR
argmax
gt coord
(c) standard coordinate decoding
(d) progressive coordinate decoding
x_1, y_1
x_2, y_2
x_k, y_k